AI Interview Simulator

An Adaptive, Voice-Driven Framework Using Dynamic CV Analysis and Retrieval-Augmented Evaluation

Abstract

Traditional automated interview systems rely on static question-and-answer datasets, limiting their capacity to assess candidates based on individualized project experience and technical depth. This paper presents the **AI Interview Simulator**, a voice-driven end-to-end framework designed for realistic, candidate-tailored technical evaluations. The system integrates a CV Parsing Engine that extracts structured candidate profiles (technical stack, domain experience, and key projects) to dynamically generate context-aware interview questions via Large Language Model APIs. Rather than using fixed answer keys, the system repurposes a hybrid Retrieval-Augmented Generation (RAG) pipeline—combining vector embeddings (ChromaDB), keyword indexing (BM25), and Cross-Encoder reranking—to evaluate candidate audio responses against domain-verified reference material in real time. Deployed through a hands-free Gradio voice interface, the platform generates comprehensive diagnostic reports and automated PDF session summaries, demonstrating a scalable approach to personalized, hands-free technical assessment.

1. Motivation

Technical hiring requires assessing candidates accurately while managing limited engineering time. Human-led technical interviews provide deep, candidate-specific insights, but they are time-consuming, expensive, and hard to scale.

Existing automated interview software relies heavily on rigid, static question banks and multiple-choice tests. These traditional setups fail to evaluate how a candidate thinks through real-world technical problems or how they execute architectural decisions based on their actual background.

The motivation behind the **AI Interview Simulator** is to create an automated, scalable interview platform that acts like a senior human interviewer. By combining Large Language Models (LLMs), dynamic CV parsing, and voice processing, the system

provides candidate-tailored technical evaluations without requiring human engineering hours for every screening round.

2. Problem Statement

Current automated technical assessment tools face three major limitations:

- **Generic, Non-Personalized Questioning:** Systems draw questions from fixed Q&A datasets. They do not adapt to the candidate's actual resume, past projects, skill level, or specific tech stack.
- **Rigid Answer Evaluation & Hallucination Risks:** Evaluating candidate responses against hardcoded answers penalizes valid alternative explanations. Conversely, relying purely on an ungrounded LLM risks accepting incorrect technical claims or hallucinated facts as correct.
- **High Interface Friction:** Text-based chat prompts fail to replicate the real-time verbal communication and pressure of an actual technical interview.

3. The Proposed Solution

The **AI Interview Simulator** provides a hands-free, voice-driven assessment pipeline that dynamically adapts to each candidate:

- **Dynamic CV-Driven Question Generation:** Replaces static Q&A datasets by parsing the candidate's PDF resume into a structured JSON schema. The system uses the OpenAI API to dynamically generate targeted technical, architectural, and project-specific questions based on the candidate's reported experience.
- **Hybrid RAG Grounded Evaluation Engine:** Converts the RAG pipeline (ChromaDB + BM25 + Cross-Encoder reranking) into a real-time evaluator. Instead of selecting questions, RAG fetches ground-truth technical documentation to verify candidate responses, detect hallucinations, and calculate technical accuracy scores.
- **Hands-Free Conversational Voice Interface:** Integrates OpenAI Whisper (Speech-to-Text) and Text-to-Speech synthesis within a Gradio interface, creating a natural, hands-free spoken interview experience.
- **Automated Analytics & Reporting:** Produces detailed session logs and exports structured PDF reports directly to Google Drive for persistent documentation.

Key Contributions

To address these challenges, the **AI Interview Simulator** introduces the following contributions:

- **Dynamic CV-to-Question Generation:** Eliminates static Q&A datasets by parsing candidate resumes into structured schemas to generate tailored technical and situational questions.
- **RAG for Real-Time Grounded Evaluation:** Repurposes hybrid RAG (ChromaDB + BM25 + Cross-Encoder) from question selection to real-time response verification, hallucination detection, and technical accuracy scoring.
- **Hands-Free Voice Interface:** Integrates speech recognition and synthesis within a Gradio framework to emulate a natural, conversational interview environment.

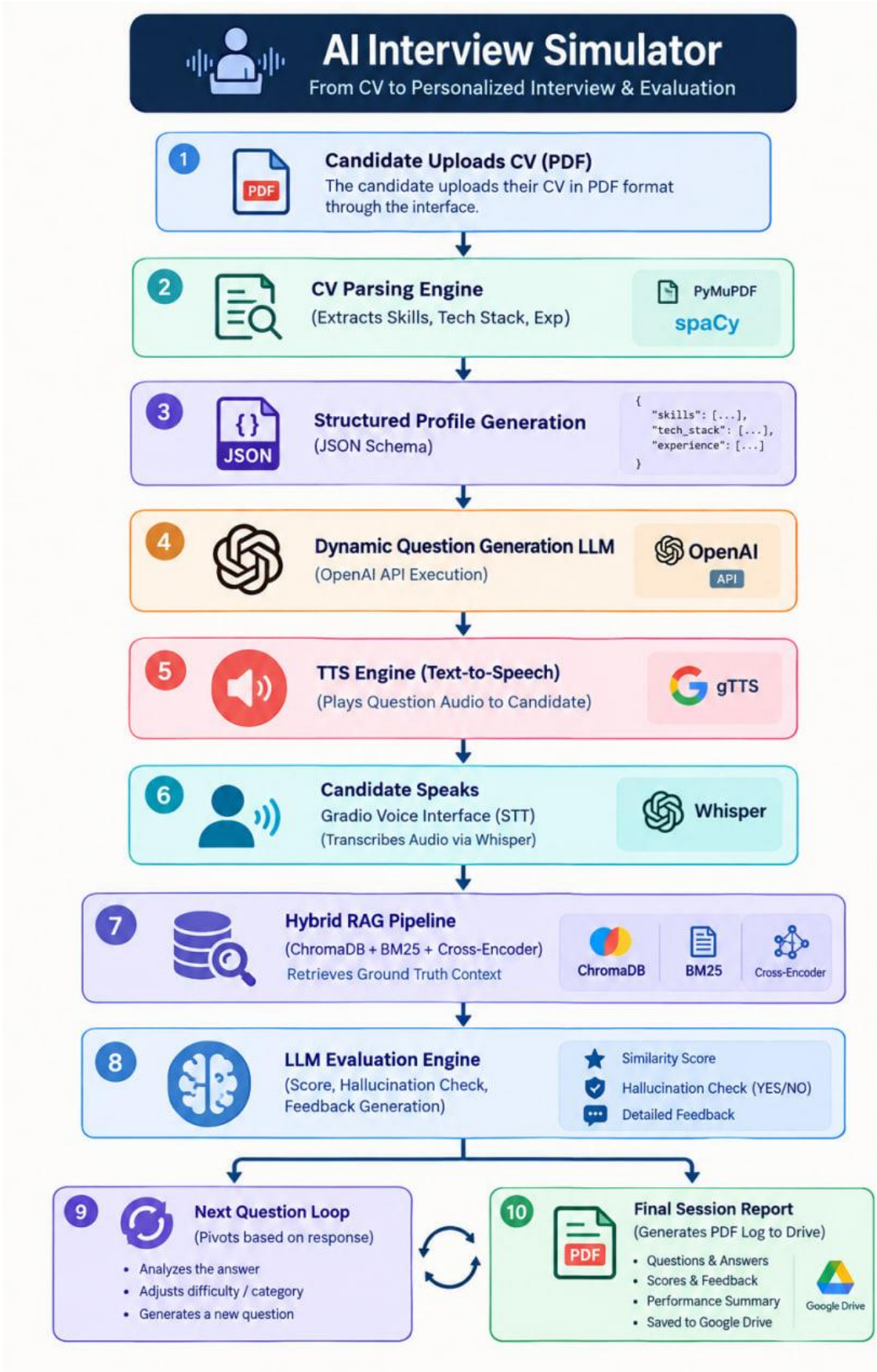
- **Project Highlights:** Key architectural decisions, methodologies, and claimed domain outputs.

1. **Problem-Solving & Trade-offs:** Scenario-based questions that test how the candidate handles system trade-offs or debugging.

- Gradio interface.
- **Audio Input:** The candidate responds verbally. The incoming audio stream is processed by OpenAI Whisper (Speech-to-Text) to generate an accurate text transcript for analysis.

Instead of relying solely on an LLM's parametric memory (which can hallucinate or accept plausible-sounding incorrect answers), the system routes candidate transcripts through a hybrid Retrieval-Augmented Generation (RAG) engine:

- **Dense Retrieval (ChromaDB):** Retrieves semantically relevant technical documentation, standard guidelines, and solution patterns based on vector embeddings.
- **Sparse Retrieval (BM25):** Ensures precise keyword matching for domain-specific terminology, library functions, and syntax.
- **Reranking (Cross-Encoder):** Scores retrieved context chunks to pass only the most accurate technical references to the evaluation LLM.
- **Scoring & Feedback:** The evaluation LLM compares the candidate's transcript against the retrieved ground-truth context to calculate a technical accuracy score, flag hallucinations, and generate actionable feedback.


AI Interview Simulator
From CV to Personalized Interview & Evaluation
1
PDF
Candidate Uploads CV (PDF)
The candidate uploads their CV in PDF format through the interface.
2
CV Parsing Engine
(Extracts Skills, Tech Stack, Exp)
PyMuPDF
spaCy
3
{ }
JSON
Structured Profile Generation
(JSON Schema)
{
"skills": [...],
"tech_stack": [...],
"experience": [...]
}
4
Dynamic Question Generation LLM
(OpenAI API Execution)
OpenAI
API
5
TTS Engine (Text-to-Speech)
(Plays Question Audio to Candidate)
gTTS
6
Candidate Speaks
Gradio Voice Interface (STT)
(Transcribes Audio via Whisper)
Whisper
7
Hybrid RAG Pipeline
(ChromaDB + BM25 + Cross-Encoder)
Retrieves Ground Truth Context
ChromaDB
BM25
Cross-Encoder
8
LLM Evaluation Engine
(Score, Hallucination Check, Feedback Generation)
Similarity Score
Hallucination Check (YES/NO)
Detailed Feedback
9
Next Question Loop
(Pivots based on response)
Analyzes the answer
Adjusts difficulty / category
Generates a new question
10
PDF
Final Session Report
(Generates PDF Log to Drive)
Questions & Answers
Scores & Feedback
Performance Summary
Saved to Google Drive
Google Drive

Related Work

Automated and voice-driven interview simulation has seen rapid advancements, transitioning from static text-based evaluations to dynamic, multimodal architectures. Recent literature highlights various approaches to automated screening, skill assessment, and performance feedback:

- **Voice-Driven Interview Simulations:** Recent frameworks, such as *AI Mock Interview: Voice-Driven Interview Simulation (2026)*, incorporate voice-based interactions and speech models (e.g., Whisper) to automate answer evaluation and generate PDF feedback reports. However, these systems often rely heavily on external AI dependencies with limited quantitative validation.

- **Multimodal and Interactive Virtual Interviewers:** Studies exploring *Virtual Interviewers, Real Results (2025)* utilized multimodal AI on computing student cohorts to provide real-time feedback during technical problem-solving and whiteboarding. While participants showed improved confidence and articulation, findings were generally restricted by small sample sizes and qualitative evaluation metrics.

- **Job Simulation & Performance Quantification Platforms:** Platforms focusing on technical and behavioral evaluation (e.g., *AI-Powered Job Simulation Mock Interview Platform (2025)*) leverage convolutional neural networks and audio analysis, though they face reliability and fairness concerns regarding automated grading. Similarly, frameworks like *AI Simulator that Quantifies Interview Performance (2026)* process audio-visual data via models such as LLaMa, Sentence-BERT, and DeepFace to achieve average response latencies of approximately 1.83 seconds, though they remain susceptible to environmental noise and lighting variations.

- **NLP and Lexical Matching Constraints:** Early text-centric approaches, such as *AI Powered Job Interview Simulator Through NLP (2026)*, utilized TF-IDF, cosine similarity, and speech recognition to compare candidate responses against reference answers. A primary limitation of these legacy models is that surface-level lexical similarity often fails to capture true semantic correctness or deep architectural reasoning.

Positioning of This Work

Unlike previous systems constrained by static question banks, pure lexical matching, or heavy reliance on external ungrounded LLMs, the **AI Interview Simulator** bridges these research gaps by integrating **dynamic CV profile parsing** with a **hybrid RAG pipeline (ChromaDB, BM25, and Cross-Encoder)**. This ensures that every generated question is tailored to the candidate's specific background, and every response is rigorously evaluated against domain-verified technical documentation to prevent hallucinations in real time.

LITERATURE REVIEW

Related Work Overview



Paper – Year	Dataset / Participants	Model / Methods	Key Results	Limitations
AI Mock Interview: Voice-Driven Interview Simulation (2026)	• Interview questions and candidate responses • Résumé-based inputs	• Gemini AI • Whisper • NLP	• Voice-based interviews • Automated answer evaluation • PDF feedback reports	• External AI dependency • Limited quantitative validation
Virtual Interviewers, Real Results (2025)	• 20 computing students	• Multimodal AI • Technical problem-solving • Whiteboarding • Real-time feedback	• Improved confidence and articulation (participant feedback)	• Small sample size • Qualitative findings • Long-term impact not established
AI-Powered Job Simulation Mock Interview Platform (2025)	• Interview responses • Multimodal inputs	• LLM • NLP • CNN • MediaPipe • Audio analysis	• Technical and behavioral evaluation • Multimodal analysis	• Limited quantitative validation • Potential reliability and fairness concerns
AI Simulator that Quantifies Interview Performance (2026)	• Candidate responses • Audio-visual interview data	• LLaMA • Sentence-BERT • DeepFace • MediaPipe • Librosa • XGBoost	• Multimodal performance scoring • Average response latency: ~1.83 seconds	• Sensitive to lighting and noise • Scoring accuracy requires further validation
AI Powered Job Interview Simulator Through NLP (2026)	• Interview questions • Reference answers • Candidate responses	• NLP • TF-IDF • Cosine Similarity • Speech recognition	• Automated response comparison • Interview performance feedback	• Incomplete experimental results • Lexical similarity may not capture semantic correctness